The Evolving Impact of Founders on Startup Employee Retention

by

Minjae Kim
Yale University

J. Daniel Kim
University of Pennsylvania

CES 26-21 **March 2026**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

Founders are known to attract prospective employees by signaling their startup's mission, culture, and potential. But do they also shape who stays? And if so, does the founder's influence diminish as the startup matures? Using matched employer-employee data from the U.S. Census, we address these questions, especially focusing on cases of founder premature death to identify plausibly exogenous exits. We find that founder departures significantly increase employee turnover. These effects are stronger in older and larger startups. Further analyses show that the impact of founder departure is more salient among employees who had longer shared tenure or have the same sex as the founder. These patterns suggest that employees develop complementarities with founders over time—an alignment in skills, relationships, or culture—that reinforce founders' influence as startups mature.

Keyword: Entrepreneurship, Employee Retention, Founders

[*] Authors are grateful for helpful comments from Diane Burton, Melody Chang, Santiago Campero, Florence Honore, David Hsu, and Ezra Zuckerman Sivan as well as various participants during seminars and conference presentations at Columbia University, HEC Paris, University of Maryland, MIT, Wharton, Duke Strategy Conference, and People and Organizations Conference. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (project 7508669: CBDRB-FY24-CED006-0021, CBDRB-FY25-CED006-010). The analysis was conducted while J. Daniel Kim was a part-time employee at the U.S. Census Bureau.

INTRODUCTION

People often pursue positions in startups for reasons distinct from those that draw them to established firms (Fairlie and Miranda 2017; Kim 2018; Sorenson et al. 2021). Many are motivated by the autonomy and agility that startups offer (Sørensen and Fassiotto 2011; Roach and Sauermann 2015). Others are drawn by the high-risk, high-reward nature of equity compensation, where substantial upside may justify early-stage volatility (Zenger 1994). It may also be the cultural or mission-driven aspects that startups often emphasize, such as breaking into new markets or solving big problems (Baron and Hannan 2002; Rocha and Brymer 2024). When jobseekers search for such qualities that are often hard to assess directly in early-stage firms, a growing body of work highlights founders as key signals. Because founders are typically the most visible leaders in young ventures, prospective employees often infer a venture's underlying values, quality, and trajectory based on the founders (Carnahan and Greenwood 2018; Campero and Kacperczyk 2020; Snellman and Younkin 2021). Accordingly, attributes such as founders' background, network, or identity shape who applies to and joins a startup (Abraham and Burbano 2022; Chung and Parker 2023; Lyttelton, Henriksen, and Sorenson 2024).

Yet while recent research has emphasized founders' outsized role in attracting talent, it remains unclear whether their influence extends beyond the hiring stage to affect employees' decisions to stay. In young firms, founders often shape not only everyday tasks and routines (Aldrich and Cliff 2003; Aldrich and Kim 2007; Beckman and Burton 2008; Marx and Timmermans 2017) but also the startup's broader strategic vision (Ding 2011; Zuzul and Tripsas 2020). These roles suggest that founders could be central to the employee experience, and, by extension, to retention. But despite this possibility, direct evidence on whether and how founders shape the employee experience and retention remains limited.

In particular, beyond the baseline expectation that founders may influence employees' decisions to stay, there are two competing sets of perspectives on how this influence may change as startups mature; and addressing these contrasting implications may offer insight into the significance of founders in startups in the long run. On the one hand, founders' influence on employees may diminish over time. A salient possibility is that, while founders may initially catalyze employees' attachment to the firm, founders become a less integral part of the organization over time, to the latter of which employees may gradually shift their loyalty (Whyte 1956; O'Reilly, Chatman, and Caldwell 1991). Thus, even if founders continue to play a role in institutionalizing those organizations and getting them "infused with value" over time (Selznick 1957; also, Stinchcombe 1997), the role may become smaller and employees' allegiance and commitment to individual leaders such as founders may wane as organizations mature.

But on the other hand, the opposite possibility is that founder influence strengthens as startups mature. Insofar as founders may be pivotal coordinators who shape employee experience (e.g., Eesley, Hsu, and Roberts 2014; Yalcinkaya, Ding, and Gupta 2025), startup employees may develop founder-specific complementarities—that is, skilled-based as well as interpersonal alignment with the founder that deepen over time. In particular, such complementarities may arise because founders are at the center of how some startups are organized. For instance, consider the case of "eponymous entrepreneurship" (Belenzon et al. 2017; Guzman and Stern 2020), where the firm's development, routines, and brand are deeply intertwined with the founder. These perspectives imply that, as startups mature, founders become increasingly central to employees' decision to stay.

This tension has both theoretical and practical significance, especially for early-stage firms already grappling with high turnover. Startups often face the well-known "liability of

newness" (Stinchcombe 1965), wherein their limited resources, underdeveloped structures, and lack of reputation may lead to disproportionately high employee attrition (Freeman, Carroll, and Hannan 1983; cf., Baron, Hannan, and Burton 2001). In parallel, turnover may be especially disruptive for startups when each employee represents a substantial portion of the firm's capacity (Ganco, Honoré, and Raffiee 2019). Thus during these formative stages when routines and cultures in the organizations are still being shaped, retaining talent is crucial (Aldrich and Yang 2014). Seen in this light, founder departure could significantly heighten this challenge, if they are key to employee retention. For illustration, consider the case of Instagram, which was acquired by Facebook in 2012 and whose founders—Kevin Systrom and Mike Krieger—stayed following the acquisition. When the founders departed in 2018, dozens of employees followed (Chozick 2020). This anecdote suggests that employees' attachment to the founders may persist beyond initial recruitment and influence their continued engagement with the firm. But at the same time, the founders' departure might simply coincide with internal shifts in mission or culture, so it is ambiguous whether these employees were reacting to the departure of the founders or something else (cf., Botelho, Gulati, and Sorenson 2024).

This tension also resonates with a well-known "horse vs. jockey" debate, which is traditionally about whether the venture's qualities (the "horse") or the founder's individual attributes (the "jockey") more strongly determine outcomes such as investment decisions, venture performance, and survival (Macmillan, Siegel, and Narasimha 1985; Gompers et al. 2010, 2020). While this debate is often framed from the perspective of investors (e.g., Kaplan, Sensoy, and Strömberg 2009), these contrasting logics imply different motivations from the vantage point of employees on whether or not to remain in firms. Addressing this debate,

therefore, may help identify distinct vulnerabilities if and when founders depart and address how much startups' "liability of newness" (Stinchcombe 1965) depends on founders.

Nevertheless, these questions present both empirical and conceptual challenges. Founders' influence often overlaps with other forces—such as investor dissatisfaction and declining firm performance—that may also shape the employee experience and retention (cf., Chen and Thompson 2015; Ewens and Marx 2018; Kulchina and Gjerløv-Juel 2019). Beyond this familiar concern, some founders may begin to gradually disengage from their firms before formally departing, making it difficult to determine whether employees leave in response to the founder's actual exit or earlier shifts in the organization. Consequently, identifying a causal effect of founder turnover on employee retention requires a setting in which founders' exit is sudden and plausibly exogenous, even though most founder departure is likely not (Valentine and Waguespack 2025).

In this article, we take on these challenges by leveraging comprehensive, administrative matched employer-employee data from the US Census Bureau between 1990 and 2014. We first document that, in a near-population of startups experiencing founder turnover (whether voluntary, investor-driven, or something else) within their first five years, employee turnover rates increase notably after the founder's exit. But given the empirical and conceptual reasons, we further exploit cases of founder premature death to identify quasi-exogenous founder turnover events (Azoulay, Graff Zivin, and Wang 2010; Choi et al. 2023; Ewens and Sosyura 2023). Using a difference-in-differences framework that compares organizations losing a founder to otherwise similar "twin" startups where the founder remains, we find consistent evidence that the loss of a founder leads to an increase in the rate of employee turnover.

We then examine how this effect varies with the maturity of the firm. In short, we find that the negative impact on retention is greater in more mature startups as measured by firm age or size. These patterns suggest that the founder's influence on employee retention seems to strengthen, not diminish, over time. Beyond these firm-level patterns, individual-level analyses further point to the salience of complementarities between founders and employees as a driver of such dynamic founder influence, as the effects are more pronounced for (a) employees with longer shared tenure with the founder and (b) employees with the same sex as the founder. In sum, founders seem to not only shape their organizations but also act as a critical anchor whose influence on employees may become deeper over time. Our findings reinforce the importance of examining how and why founders actively anchor employees' attachment—an area critical to startups' ability to survive and scale amidst their inherent liabilities of newness.

THEORETICAL FRAMEWORK

Significance of Founders for Prospective Employees' Decisions

Startups often lack—relative to their established counterparts—financial resources, industry networks, routines, and reputation that often facilitate more effective and efficient human resource management (Stinchcombe 1965; Cardon and Stevens 2004; Beckman and Burton 2008; Wasserman 2012; Fairlie and Miranda 2017; Honoré and Ganco 2023). But despite these organizational shortcomings, many individuals choose to join startup employers, partly because they are motivated by what those startups will become and not necessarily what the startups represent at the time of joining (cf., Kim 2018; Sorenson et al. 2021). This suggests that, beyond general motives for joining firms, prospective employees may look for cues that the startup possesses both the capacity and intent to fulfill its stated mission over time (cf., Hannan and

Freeman 1984). But startups still face the "liability of newness" (Stinchcombe 1965), based on which they may fall short of signaling such prospects. After all, startups lack a proven track record or credible signals of their capability or commitment to carry out their stated mission.

To this, recent work has highlighted the role of founders (Carnahan and Greenwood 2018; Campero and Kacperczyk 2020; Snellman and Younkin 2021; Abraham and Burbano 2022; Chung and Parker 2023; Lyttelton et al. 2024; Rocha and Brymer 2024). The idea here is that, insofar as founders are arguably among the most important leaders in the nascent firms, prospective employees gather signals about the startups' underlying qualities from founders and are drawn to the organizations based on founder characteristics. For instance, job applicants may find founders' sex as a signal of certain quality and values (Campero and Kacperczyk 2020) and sort themselves into startups whose founders seem to reflect the firms' stated mission more faithfully (Abraham and Burbano 2022; cf., Carnahan and Greenwood 2018). Applicants may also infer the startups' culture and quality based on founders' educational background (Chung and Parker 2023) or demographic characteristics (Snellman and Younkin 2021).

The significance of founders in employees' decision to join may also be based on social networks. Above and beyond the growing evidence that social relationships provide critical conduits of coordination for founders to access ideas, funding, suppliers, and customers (Shane and Stuart 2002; Ruef, Aldrich, and Carter 2003; Stuart and Ding 2006; Dahl and Sorenson 2009), there also is evidence that social relationships provide important conduits of information to (potential) employees, as a large share of startup employees seem to come from the pool of the founders' families, friends, and prior business affiliates (Ruef 2010). Most recently, Lyttelton et al. (2024) and Rocha and Brymer (2024) document that startups are much more likely to hire as employees those with prior relationships to the founders. Their evidence is especially notable in

being able to identify jobseekers who likely have relationships with the founders but could have joined other firms.

These studies in sum identify a hiring process that may be unique to startups, whereby founders may play a large role. Naturally, there are two related implications. First, insofar as employees may join startups for the greater autonomy and nimble environments, high-risk high-reward structure, and/or novel ideas and missions that startups promise but are difficult to directly observe in the present, they may look to founders as convincing signals of those underlying qualities. To be sure, this does not imply that founders are the only—or even the most salient—such signal, but they may be a salient one, especially when prospective employees and others may look to founders as leaders shaping the firms (cf., Selznick 1957).

But more importantly, second, the fact that employees consider founders as indicative of the firm's underlying qualities when initially joining may also suggest that they see founders as a critical factor in their decisions to stay or leave *after* joining the firm. This idea stems in part from Schneider (1987) attraction-selection-attrition (ASA) model: Organizations are a function of the types of people they initially attract and people who select into them; therefore, employee attrition depends partly on what attracted those employees into organizations in the first place. Again, employees' decisions to stay or leave likely depend on a diverse set of factors, including personal ones. But the idea is: Insofar as employees make their decisions to join at least partly based on founders, they may also make their decisions to stay or leave partly based on them. This latter implication therefore sets up the baseline expectation that employees may make their decisions to stay or leave firms depending on the founder.

Do Founders' Influence on Employee Retention Diminish as Startups Mature?

Yet, while the foregoing logic for founders' overall influence on current employees' decisions to stay may be relatively straightforward, it is less clear how founders' influence on employee retention will change as their startups mature. This question may have straightforward practical import: Employee retention may be more difficult in the event of founder departure depending on where firms are in their lifecycle. This question also may help address from where such founder influence on employees comes. However, there seem to be two competing implications from existing arguments on founders' roles in startups over time.

On the one hand, it seems plausible that such founder influence diminishes as the firm matures. A primary premise here is that, as the firm matures, it may also develop its own structure and routines increasingly independently from founders (cf., Hannan and Freeman 1984). If this is the case, employees may increasingly rely less on founders and more on the organization itself. As a consequence, employees may also increasingly make their decisions to stay or leave their startups independently from founders. To be sure, this does not imply that founders become irrelevant in more mature firms—even as structures and routines evolve in ways that may reduce their direct involvement. That is, employees may still look to founders for further institutionalization of those organizations becoming "infused with value" over time (Selznick 1957; for a more recent revisit to this logic, see King 2015). But this does suggest that employees may rely on founders *less* as the firm matures, insofar as the startup's maturity may imply structural independence from founders' idiosyncrasies (Stinchcombe 1997).

This implication consequently articulates one possible extension of some foundational ideas around organizational development (Stinchcombe 1965; Hannan and Freeman 1984; O'Reilly et al. 1991), where the organization becomes increasingly independent from founders as they professionalize, and in turn, employees' decisions to stay in the organization also develops

increasingly independent from founders over time. This perspective thereby suggests that the object to which employees become attached is the organization, and founders' role in facilitating that attachment process diminishes over time. Consequently, as the firm matures, a natural outcome seems to be the "organization man" (Whyte 1956), whose allegiance and commitment is increasingly toward organizations and not any single individual therein (cf., Srivastava et al. 2018).

Yet on the other hand, an equally plausible but contrasting implication is that founder influence on current employees grows as firms mature. One possible reason is based on complementarities that founders and employees develop with one another. In particular, when startups often require roles that are defined contingently upon unique combination of employees (Burton and Beckman 2007; Beckman and Burton 2008) and tasks that are completed through collaboration (Aldrich and Cliff 2003; Aldrich and Kim 2007; Jaravel, Petkova, and Bell 2018), employees may develop skill-based, interpersonal, and/or other types of complementarities with founders. Recent findings from careful empirical examination indeed suggest that individuals in teams often develop "team-specific capital" (Jaravel et al. 2018; see also Eesley et al. 2014 and Baek, Bidwell, and Keller 2022), whereby their value depends on their unique complementarities with others in the same team. Further, these complementarities may go above and beyond such economic considerations, as employees may develop social relationships that hold one another down in the same organization (Castilla 2005; cf., Marsden and Campbell 1984, 2012; Burt 2002; Kleinbaum 2012, 2018; Zhang, Aven, and Kleinbaum 2024; Aven 2015; Pillemer and Rothbard 2018; Kim and Fernandez 2023). In other words, such complementarities may be not only skill- and task-based, but also cultural and interpersonal.

This idea becomes especially salient in situations where founders become more important elements of the organization's structure and routines over time. For instance, prior research has shown that founders' professional-educational background can exert lasting influence on firm strategy, especially in shaping practices that persist beyond the founding stage (Ding 2011). The growing influence of founders seems especially likely in startups whose blueprints emphasize the role of their founders, making the organization's development dependent on these central individuals (Baron and Hannan 2002). This is perhaps best illustrated in "eponymous entrepreneurship" (Belenzon et al. 2017; Guzman and Stern 2020), where the identity and routines of the firm are explicitly tied to the founder. In this case, as the startup evolves, the founder's role often expands in parallel, transforming employees' attachment to the organization to become increasingly mediated through their connection to the founder, given that the organization develops its structure and routines around the founder.

This set of accounts on founders' role hence implies a contrasting dynamic (i.e., over time) implication—that founder influence on employees grows as the firm matures. This implication stems in part from the complementarities that employees develop specifically with founders, which may arise particularly in situations where founders become more central in organizational structure and routines over time. In sum, these accounts imply that founders may impact employee turnover more as the organization becomes more mature.

Consequently, our foregoing discussion results in two competing implications. On the one hand, founders' significance on employee turnover may diminish as the firms mature, based on a plausible premise that founders' role becomes less salient with the organizations' development; on the other hand, the significance of founders' role for employees' decisions to stay may intensify as the firms mature, based on employees' growing complementarities with

their founders and/or organizations that develop around their founders. Since these two possibilities are based on different sets of accounts on why founders matter to employees, examination of founder influence on employee turnover probes the degree to which either set of arguments—albeit not necessarily specific explanations within each set—may account for the empirical outcome of founders' presence. Therefore, our empirical tests below are by design exploratory. Yet, they serve as a useful opportunity to not only theoretically identify the nature and extent of founder influence on current employees but also practically point to salient determinants of employee turnover that is especially pertinent to young firms.

DATA AND SETTING

Sample Construction

For our empirical examination of the two competing implications, we use administrative data of employers and employees from the U.S. Census Bureau. The primary dataset is the Longitudinal Business Database (LBD), which is a panel dataset of all establishments in the U.S. with at least one paid employee (Chow et al. 2021). This dataset covers years 1976-2021 and all industries in the private non-farm economy across all U.S. states. Given the longitudinal nature of the LBD, we can identify the birth and exit of startups and measure various business characteristics such as employment size, payroll, and firm age.

The other dataset used in this study is the Longitudinal Employer-Household Dynamics (LEHD), which is a panel dataset of workers covering roughly 95% of private sector jobs in the U.S. We use the full available version of the LEHD which covers all U.S. states except Massachusetts. While LEHD's time coverage spans from 1985 to 2014, states vary in when they

first appear in the LEHD depending on their initial participation in the LEHD program. Most states are available after 2000 (Abowd, Haltiwanger, and Lane 2004).

We integrate these two databases to construct a sample of startups and their employees including founders. From the LBD, we select startups that are five years old or younger which is based on the firm's initial year of positive payroll (i.e., employ at least one paid worker). Then, we use the LEHD to identify all paid employees associated with each startup. Among all wage earners, we identify founders following a standard approach used in previous Census-based studies (e.g., Kerr and Kerr 2016; Kim 2024). Specifically, founders are individuals who are employed beginning in the first year of the firm and among the top three earners in the firm during the first year. The remaining individuals are labeled as non-founding employees.

Based on this empirical apparatus, we begin with a broad sample of all startups that experience the departure of a founder. Founder departure is defined as when a founder no longer earns wages from the firm for any reason ranging from retirement to dismissal. We identify roughly 135,000 startups that experience a founder departure—before the firm's age of ten—between 2000 and 2015. Then, we construct a panel sample of these firms including each of the five years (if available) before and after the year of the founder departure. This starting sample allows us to assess whether and how rates of employee turnover generally change after a founder separates from the startup.

Causal Identification

An observed relationship between founder departure and employee turnover in a sample constructed as aforementioned is not necessarily causal in nature. One potential confounder is the (unobserved) changing quality of the startup. For instance, a major technological setback in a

startup may lead to increases in both the risk of founder departure and the rates of employee turnover. Another potential concern is gradual withdrawal, whereby founder exist may not necessarily coincide with other factors but it happens slowly over time such that it is difficult to pinpoint at what point employee experience was caused by founder exit. These concerns therefore set up a trade-off in this causal identification, where observed founder exits may not necessarily resemble the most "typical" founder exit events but that may help us rule out effects of those confounders and gradual withdrawal (Valentine and Waguespack 2025).

An ideal experiment is to randomly remove the founder from some startups while leaving others intact. While an empirical design that exactly maps on to this idea is not feasible, we leverage premature death among founders as a natural experiment, which prior studies have used to identify the effects of human capital departure (Azoulay et al. 2010; Carnahan 2017; Choi et al. 2023; Ewens and Sosyura 2023; Jaravel et al. 2018). Following the approach of Jaravel et al. (2018), we define premature death as one that occurs before the age of 60 based on the date of death using the Numerical Identification File at the Census Bureau. This approach allows us to isolate a sample of startups that experience a quasi-exogenous loss of a founder (i.e., "treated group"). One limitation worth noting is that we cannot observe the nature or cause of death—in contrast to Azoulay et al. 2010 and Ewens and Sosyura (2023) who manually track the cause of death—because our data do not contain individual names or social security numbers. Nonetheless, to help ensure that the events are sudden and unexpected (as opposed to following a long spell of illness or precursors), we restrict to events in which the individual receives regular wages in the quarter during which the death occurs. In other words, individuals who experience an unemployment spell or a continued decrease in wages prior to the quarter of their death are not included in our sample.

In addition, we construct a control group of startups that are observationally similar to the treated group but do not experience founder loss. We do so by coarsened exact matching on the following characteristics as measured in the year prior to the death shock: firm size, firm birth year, NAICS-4 industry code, and state. We pair each treated startup with one control startup, resulting in a sample of approximately 16,000 treated startups and 16,000 control startups.

Methodological approach

To estimate the effect of founder departure on employee turnover rates for startup *j* in year *t*, industry *n*, and state *s*, we use the following regression equation with leads and lags around the premature death of the founder:

$$RateTurnover_{j,t} = \gamma_t + \alpha_n + \delta_s + \rho_a + \sum_{k=-5}^{5} \beta_k d[k]_{j,t} \times Treated_j + \sum_{k=-5}^{5} \phi_k d[k]_{j,t} + \varepsilon_{j,t}$$

where the main dependent variable $RateTurnover_{jt}$ is the percent share of individuals employed at firm *j* in year *t* who no longer earns wages from the focal firm in year *t+1* and becomes employed at another employer for high wages. Because the LEHD does not distinguish voluntary from involuntary turnover, this additional restriction based on wage-increasing turnover allows for an interpretation closer to voluntary turnover, as prior studies show that voluntary (involuntary) turnover is systematically associated with higher (lower) wages (e.g., Keith and McWilliams 1995).[2] Furthermore, we measure $d[k]_{j,t}$ as a binary variable equaling 1 if year *t* is *k* year—ranging from -5 to 5—from the year of the founder death shock. For the control group, which by definition does not lose a founder, *k* is defined relative to a "placebo" year (i.e., the

[2] Despite this attempt, one limitation is that we ultimately cannot directly observe whether a turnover is voluntary or involuntary. Therefore, we conduct robustness checks of measuring turnover rates without this condition of a wage increase and find substantively similar patterns.

year at which its treated counterpart experiences the founder loss). We include fixed effects for the NAICS-4 industry (α_n), state (δ_s), year (γ_t), as well as the age at which the startup experiences the death (placebo) shock (ρ_a). We cluster standard errors $\varepsilon_{j,t}$ at the firm level. In some of our analysis, we provide difference-in-differences estimates by defining $Post_{j,t}$ as a binary variable equaling 1 if *k* is between 1 and 5, and 0 otherwise.

It is worth noting that our sample of firm-year observations is an unbalanced panel. The first reason is left-truncation resulting from the timing of firm birth and founder loss. For example, a startup that experiences founder loss at age four did not exist in year *t-5* relative to the founder death shock. Such observations are dropped from the sample. The other reason is right-truncation similarly resulting from the timing of founder loss and founder loss. Here, a startup that exits (e.g., firm death or acquisition) in year *t+4* since the founder death shock will not exist in year *t+5*. These post-exit observations are also dropped. This approach ensures that firm death is not observed with a rate of turnover of 100%. Our unbalanced panel set contains about 221,000 observations.

RESULTS

Table 1 provides summary statistics of the firms in our study. As expected, the startups in our sample are (by construction) young with an average of about 3 and small with an average employment size of about 7. Given that we use administrative data from U.S. Census that covers the near-population of startups, our sample consists of both small business and growth-oriented startups (cf., Botelho, Fehder, and Hochberg 2021). As a proxy for this distinction, we categorize whether the startup operates in a high-tech industry, which is classified by Goldschlag and Miranda (2020) based on industries with the highest shares of STEM-related workers. About 7%

of our sample are in high-tech industries. In terms of geographic variation, California, Texas, and Florida are the most heavily represented states. The average startup has revenues of slightly below $1 million, though there is significant variation.

[Insert Table 1 here]

Founder Departure and Rates of Employee Turnover

We begin our analysis of employee turnover using a broad sample of all startups that experience a founder departure for any reason ("separation"). As shown in Specification 1 of Table 2, we observe that employee turnover rates are significantly higher following the separation of a founder. This association is similarly positive after accounting for the startup's industry and state in Specification 2 as well as the firm age at which the separation occurs in Specification 3. In addition to the statistical precision of the coefficients at the $p<0.01$ level, the economic magnitudes are sizeable. Relative to the baseline rates of employee turnover prior to founder separation, this estimate in Specification 3 represents an increase of roughly 27%.

[Insert Table 2 here]

However, these patterns are difficult to interpret because of potential confounders. As discussed earlier, one possibility is that the declining (unobserved) quality of the startup may catalyze both the founder's decision to leave and employee turnover rates. Another concern is that employee turnover rates generally rise as a startup matures, leading to a mechanical increase associated with the post-founder departure period. To overcome these challenges, we turn our analysis to the set of startups that experience a premature death of a founder. We compare this set to a control group of similar startups that do not lose a founder, allowing for a difference-in-differences framework. Table 3 shows the results from this analysis. In Specification 1, we find a

positive estimate on *Post*, which indeed suggests that startup employee turnover becomes more probable over time. Nonetheless, we find a strongly positive effect on *Treated* × *Post*, even after including industry and state fixed effects in Specification 2 and also firm age fixed effects in Specification 3. Furthermore, we include firm fixed effects in Specification 4 to absorb all time-variant characteristics of the firm; note that *Treated* and some of the other firm-specific fixed effects are omitted due to collinearity. The estimate is substantively similar. These patterns demonstrate that employee turnover rates significantly increase following the loss of a founder compared to similar startups that do not experience such a shock. In terms of the economic magnitude, the coefficient in Specification 3 represents a 19% increase in turnover rates relative to the pre-treatment mean of the control group.

[Insert Table 3 here]

We further unpack these results in an event study design by separately estimating the effect in each year before and after the treatment. Figure 1 plots the coefficients along with their 95% confidence intervals. An identifying assumption in the difference-in-differences framework is one of parallel trends—that is, the treated group is on a similar trajectory as the control group absent the treatment itself. The estimates in the pre-treatment period are statistically indistinguishable from zero, which lends support to the underlying assumption of parallel trends. More substantially, we find that the effect of founder loss on employee turnover is swift. The effect is strongest in the first two years of the treatment and subsequently dwindles. One possible reason for this decline is that some of the employees who join *after* the founder loss are uninfluenced by death shock because they most likely did not have an opportunity to develop founder-based attachment—at least in the context of current employment—with the founder, although any conclusions based on this trend alone are preliminary.

[Insert Figure 1 here]

Why Does Founders' Influence on Employee Retention Strengthen Over Time?

Our theoretical assessment identifies two opposing predictions on how founders' influence on startup employee retention can change as startups mature. On the one hand, this effect of founder departure on employee retention may become less consequential as the organization ages, if the organization develops increasingly independent from founders and employees develop their attachment to the organization only. On the other hand, the effect of founder departure may grow stronger over time if founders directly shape the employee experience and/or become increasingly central in the organization, through developing idiosyncratic complementarities. To empirically adjudicate between these opposing implications, we examine whether the results meaningfully vary with the timing of founder departures relative to firm maturity.

Table 4 begins by splitting the sample into startups at or below the median age during which the death shock occurs (3 years old) in Specification 1 and those above the threshold in Specification 2. The estimated effect for older startups is noticeably higher than that for their younger counterparts. When tested, the postestimation difference in the two models for *Treated* × *Post* is statistically significant with a p-value of 0.014. For further robustness, we use a continuous measure of firm age in a triple interaction regression in Specification 3. The coefficient on the triple interaction term is positive and weakly significant with a p-value of 0.087. This implies that the treatment effect of losing a founder on employee turnover rates tends to increase as the firm is older during the year of the founder loss. Furthermore, we use firm size as an alternative measure for maturity in Specification 4. We similarly find a weakly positive

interaction effect with a p-value of 0.046—a pattern that further supports the idea that the founder's influence on employee retention strengthens as startups grow.

[Insert Table 4 here]

Probing the Extent to Which Employee Turnover Is Driven by Founder-Employee Complementarities

So far, these findings suggest that founders' influence on employee retention grows stronger as the firm matures. However, interpreting these results is far from clear. One difficulty is that firm age (and size) is an imperfect proxy for startup maturity. In addition, this approach potentially introduces a survival bias—that is, older firms are those that managed to survive for a longer amount of time, which may be related to the relative of importance of their founders. Thus, we transition to an individual-level analysis to get a closer view. This shift also provides the opportunity to more directly examine the complementarities between founders and employees that explain the growing importance of founders on employees.

More specifically, we probe the degree to which employees' founder-based attachment to the firm may be explained by the founder-employee complementarities in two ways: (1) how much on-the-job experience (i.e., tenure) the founder and employee share, and (2) whether the founder and employee share the same sex. First, we examine whether the treatment effect is moderated by the employee's tenure within the firm—measured as the number of years between the initial joining of the firm and the departure of the founder. In this setup, tenure serves as a proxy for the extent to which the founder and employee worked together and therefore presumably developed complementarities with one another. If founder-employee complementarities are an important driver of employee retention, the resulting prediction is that

employees who share longer tenure with the founder are more severely affected by the founder's exit compared to their colleagues who spent less time with the founder.

As tenure varies across individuals, we switch our analysis to the individual-level using a Cox hazard framework to model each person's risk of turnover. It is worth noting that this analysis excludes individuals who join after the founder departure event as these individuals by definition do not have an employment-based relationship with the (departed) founder in that firm. Control group of similar startups that do not experience founder loss are included using the "placebo" year (i.e., the year at which its treated counterpart experiences the founder loss) to calculate individuals' tenure. Summary statistics of these individuals are provided in Table 5.[3]

[Insert Table 5 here]

Table 6 displays the results from these Cox hazard regressions. As a baseline, Specification 1 shows that treated individuals are indeed associated with higher rates of turnover than turnover rates of those who do not experience founder loss. We introduce employee tenure as a key construct in Specification 2. In terms of the coefficient of interest, the estimate on *Treated* × *Tenure* is strongly positive and significant at the $p<0.01$ level. This suggests that, among the treated group, individuals who shared a greater amount of co-working experience with the founder are more likely to leave after founder premature death. This positive effect is not a reflection of long tenure itself: In fact, the baseline effect of *Tenure* in Specification 2—which represents the estimate for the control group—is negative and significant at the $p<0.01$ level. In other words, individuals in the control group with longer tenure are *less* likely to leave the firm—a finding that is consistent with prior literature on worker tenure and turnover (e.g.,

[3] As described in the section *Sample Construction,* the matching procedure to construct our sample was done at the firm-level, rather than at the individual-level. As such, summary statistics of individuals in Table 5 show small differences in the sample size and individual characteristics.

Jovanovic 1979).[4] We find substantively similar results after controlling for demographic characteristics of the individual (Specification 3) as well as year fixed effects (Specification 4) and state and industry fixed effects (Specification 5). Together, these results imply that, among the treated group, individuals who shared a greater amount of co-working experience with the founder—above and beyond their time spent in the firm per se—are more likely to leave following the loss of a founder. These findings thereby suggest that the founder and employee develop complementarities with one another, thereby reinforcing employees' retention prospects with the firm over time when the founder remains in the firm.

[Insert Table 6 here]

Our second test is based on shared demographic characteristics. In particular, the literature on sex-based homophily suggests that individuals tend to form (stronger) bonds with same-sex others (McPherson, Smith-Lovin, and Cook 2001). Prior work in entrepreneurial contexts also find such impacts of sex-based homophily, where people of the same sex seem more likely to attract and affect one another (Kacperczyk 2013; Campero and Kacperczyk 2020; Abraham and Burbano 2022). Based on this, the idea here is the same as the tenure-based analysis. That is, if founders share the same sex with employees, founder-employee complementarities may manifest more strongly, such that founder exit may have disproportionately larger impacts. If such complementarities do not exist, founder exit should not disproportionately affect same-sex employees any more than different-sex ones.

[Insert Table 7 here]

[4] For instance, Jovanovic (1979) theorizes that, as workers learn about their fit with an employer over time, those that discover an unfavorable fit may tend to leave early. By this logic, the remaining workers may be those with better fit and higher retention prospects.

Table 7 presents results from these tests. In Specification 1, we find that *Same Sex* is negatively associated with employee turnover risk. This is consistent with the notion that founder-employee complementarities—greater salience proxied by shared sex—may help reduce employees' baseline propensity to leave the firm. However, the interaction between founder departure and *Same Sex* is negative and statistically significant at the $p<0.01$ level. This result indicates that founder departures have a substantially stronger disruptive effect on employees who share the departing founder's sex, over and above the generally stabilizing effect of shared sex. The pattern remains robust in Specifications 3 and 4, which include additional controls. Consequently, it appears that sharing the same sex serves not only to deepen employee integration under normal circumstances but also to heighten the sense of disruption when the founder departs. More broadly, these findings from individual-level analyses serve as evidence that employees and founders develop complementarities with one another, thereby amplifying the turnover-inducing effect of founder separation over time.

Robustness Checks

Our identification strategy based on premature deaths among founders is appealing because of its quasi-exogenous nature—that is, employees are faced with a sudden loss of a founder. Nonetheless, one feature of this exogenous shock is that the experience of losing a colleague due to a premature death can be emotionally distressing. This, in turn, may encourage affected employees to leave the firm to distance themselves from and thereby cope with the tragic event. As such, the observed effects of founder turnover may be driven by the general emotional distress of losing a colleague rather than the theorized influence of founders. Unfortunately, we

do not observe the emotional state of employees and therefore cannot directly rule out this alternative explanation.

Instead, we conduct a "placebo" test in which we repeat the same empirical exercise but one using premature death among non-founder employees—specifically, individuals who join the company after the first year of its founding. To the extent that an unexpected passing of a colleague in and of itself causes emotional distress—regardless of whether the colleague is a founder or rank-and-file employee of the company—we should expect this effect when the startup tragically loses a member. To conduct this test, we collected a sample of startups that lose a non-founder employee due to a premature death, along with a control group of observationally similar startups as done previously with the original sample. This results in a sample of roughly 17,000 startups.

Table 8 shows the results from this analysis. As shown in Specification 1, the estimate is positive but statistically and economically indistinguishable from zero. Similarly, we fail to find a statistically or substantively significant positive effect associated with non-founder employee loss on employee turnover rates when we include fixed effects in other specifications. We further examine these estimates in an event study form. As displayed in Figure 2, there is a positive effect on employee turnover rates in the year that the non-founder employee is lost. But although this estimate is statistically significant at the $p<0.05$ level, it represents a relatively small increase of 7% relative to the baseline mean of the control group. Furthermore, we do not find any meaningful effect in any of the subsequent years. Taken together, these findings show that losing a founder has a uniquely profound impact on the startups' personnel. While we cannot definitively rule out the presence of emotional distress following the sudden loss of a colleague,

these null results lend additional support to the evidenced link specifically between founder departures and increased rates of employee turnover.

[Insert Table 8 here]

[Insert Figure 2 here]

DISCUSSION

Employee retention is a critical concern for startups, which often lack many of the resources that help more established firms retain their employees. With respect to that concern, this article shows that founders play an important role in emboldening their firms' prospects of retaining talent. In particular, our findings show that such impact likely comes from complementarities that founders and employees develop over time. These findings from the near-population of startups in the U.S. spanning 1990-2015, along with a more focused sample using founder premature deaths for causal identification, show support for the salience of such founder-employee complementarities and thus the increasing influence of founders' role over time.

Most immediately, these findings and theoretical lessons point to the dynamically growing importance of founders in human resource management of startups. Entrepreneurs often capture the public's imagination, as they are often seen as inspiring creative ideas into contemporary U.S. economy and society more broadly. But at the same time, their role as managers in organizations has not been explored as extensively. This paper identifies one important side of founders' role, where their complementarities with employees operate as a mechanism through which employees may decide to stay in the organizations. To the extent that stronger employee retention may be a foundation for future firm performance, founders' economic contributions to entrepreneurial organizations may be deeply rooted in their roles as central social actors (Baron and Hannan 2002; Zuzul and Tripsas 2020). To be sure, this by no

means necessarily implies that founders are "good" managers—more specifically, ones who are capable of helping retain employees in the organization in general. Rather, this implies that employees who initially *selected into* and subsequently stayed in the organizations alongside founders (Schneider 1987) are more likely to leave the organizations when founders leave the organizations, and even more so when they have developed deeper complementarities with those founders.

This implication may be practical and empirical as well as theoretical. In particular, firms that acquire startups often replace founders so that they can steer the startups and more particularly lead people in the organizations in directions that they would prefer (Chen and Thompson 2015; Ewens and Marx 2018). In such events, Kim (2024) shows that employee turnover after the acquisition tends to be especially high when founders leave shortly after the deal. Our findings suggest that such patterns are not selective to acquired firms, thereby proposing founders as an independent driver of employee retention for startups more generally. A flipside of this lesson is that founders may be an impediment to the firm's dynamism, if it comes partly from lack of employee turnover (c.f., Schumpeter 1976). This multifaceted role of founders with respect to their firms' dynamic consequently imply that startups may often face a trade-off when considering the founder replacement (cf., Wasserman 2012).

Our findings may furthermore apply more broadly to our understanding of organizations and their process of development more generally. A key premise in our understanding of organizations often is that as organizations mature, so do their norms, values, and routines (Hannan and Freeman 1984; Stinchcombe 1997). In turn, these organizational processes may shape employees' attachment to their employers. Therefore, employees' higher likelihood of turnover has often been thought of as a core element of startups' "liability of newness"

(Stinchcombe 1965). Our findings do not contradict such ideas and evidence. Rather, they imply that such process of organizational development may occur concurrently with the development of idiosyncratic complementarities between the founder and employees, the former of who are often perceived as exerting diminishing significance in contemporary organizations (cf., Selznick 1957, 1996; Scott 1995). That is, organizations developing as the "iron cage" and rational entities driven by "legal-rational authority" may still be places where "charismatic authority" operates saliently (Weber 1947).

In drawing this lesson, we also highlight the particular use of our causal identification. This paper is not the first one to use premature death as an exogenous shock source of variation in the departure of an individual. However, a two-fold challenge often looms large in using (founder) premature death as an exogenous shock, as it often implies removal of the individual from the organization (which is the focus of our empirical examination) *and* emotional distress that comes from such premature death (which is not). In fact, insofar as any exogenous change can instill emotional distress to some extent, this challenge looms large to this general type of identification strategy. Therefore, some of previous studies notably examine theoretical contexts where occurrence of both sudden removal and emotional distress are important (e.g., Shi, Hoskisson, and Zhang 2017; Azoulay, Wahlen, and Zuckerman Sivan 2019). However, our placebo test clarifies the bounds and scope conditions of our theoretical framework, in that non-founder employees' departure does not meaningfully affect other employees' rate of turnover. In other words, it is not the tragic departure of any individuals in the organization but specifically that of important leaders such as founders that elicits turnover of other employees. After all, not every type of complementarities are the same.

Yet, some limitations in our study are worth noting. Given that we do not observe the formal role that founders play in their organization, we do not make any claims as to how the founder-employee complementarities may affect employees' formal roles. But our findings identify founders as playing *some* roles, based on which employees develop motives to stay in the organization (cf., Wasserman 2003; Chen and Thompson 2015; Lee, Yoon, and Boivie 2020). By the same token, we also caution that founders may not be the only leaders who may play such (dynamic) roles in employee retention. Yet, by focusing on founders and not those in certain formal roles, our findings may present a more conservative estimate of the leaders' impact on employee retention, insofar as founders' formal roles may not always live up to the significance of their informal roles as founders. Nevertheless, we see it as a productive avenue for future research to explore the interactions between founders' formal and informal roles (cf., Wiersema 1992; Westphal and Fredrickson 2001; Quigley and Hambrick 2012; Karaevli and Zajac 2013).

CONCLUSION

Startups not only contribute to important economic outcomes such as job creation and innovation but also serve as an empirical laboratory to study the evolution of organizations. This paper leverages this opportunity and shows that founders as individual leaders may increase their influence on employee retention in the organizations over time, thereby overshadowing the process of organizational development as a concurrently developing influence in the domain of employee retention. An upshot is that a mix of "charismatic authority" and "legal-rational authority" (Weber 1947) may only get stronger in organizations in the long term, rather than the latter overwhelming the former over time, thereby calling for amendment to our understanding of

modern organizations as the "iron cage" with diminishing influence of individual leaders' idiosyncrasies.

REFERENCES

Abowd, John J., John Haltiwanger, and Julia Lane. 2004. "Integrated Longitudinal Employer-Employee Data for the United States." *American Economic Review* 94(2):224–29. doi:10.1257/0002828041301812.

Abraham, Mabel, and Vanessa Burbano. 2022. "Congruence Between Leadership Gender and Organizational Claims Affects the Gender Composition of the Applicant Pool: Field Experimental Evidence." *Organization Science* 33(1):393–413. doi:10.1287/orsc.2021.1442.

Aldrich, Howard E., and Jennifer E. Cliff. 2003. "The Pervasive Effects of Family on Entrepreneurship: Toward a Family Embeddedness Perspective." *Journal of Business Venturing* 18(5):573–96. doi:10.1016/S0883-9026(03)00011-9.

Aldrich, Howard E., and Phillip H. Kim. 2007. "Small Worlds, Infinite Possibilities? How Social Networks Affect Entrepreneurial Team Formation and Search." *Strategic Entrepreneurship Journal* 1(1–2):147–65. doi:10.1002/sej.8.

Aldrich, Howard E., and Tiantian Yang. 2014. "How Do Entrepreneurs Know What to Do? Learning and Organizing in New Ventures." *Journal of Evolutionary Economics* 24(1):59–82. doi:10.1007/s00191-013-0320-x.

Aven, Brandy L. 2015. "The Paradox of Corrupt Networks: An Analysis of Organizational Crime at Enron." *Organization Science* 26(4):980–96. doi:10.1287/orsc.2015.0983.

Azoulay, Pierre, Joshua S. Graff Zivin, and Jialan Wang. 2010. "Superstar Extinction." *The Quarterly Journal of Economics* 125(2):549–89. doi:10.1162/qjec.2010.125.2.549.

Azoulay, Pierre, J. Michael Wahlen, and Ezra W. Zuckerman Sivan. 2019. "Death of the Salesman but Not the Sales Force: How Interested Promotion Skews Scientific Valuation." *American Journal of Sociology* 125(3):786–845. doi:10.1086/706800.

Baek, Minseo, Matthew Bidwell, and Jr Keller. 2022. "My Manager Moved! Manager Mobility and Subordinates' Career Outcomes." *Organization Science* 33(5):1861–88. doi:10.1287/orsc.2021.1509.

Baron, James N., and Michael T. Hannan. 2002. "Organizational Blueprints for Success in High-Tech Start-Ups: Lessons from the Stanford Project on Emerging Companies." *California Management Review* 44(3):8–36. doi:10.2307/41166130.

Baron, James N., Michael T. Hannan, and M. Diane Burton. 2001. "Labor Pains: Change in Organizational Models and Employee Turnover in Young, High-Tech Firms." *American Journal of Sociology* 106(4):960–1012. doi:10.1086/320296.

Beckman, Christine M., and M. Diane Burton. 2008. "Founding the Future: Path Dependence in the Evolution of Top Management Teams from Founding to IPO." *Organization Science* 19(1):3–24. doi:10.1287/orsc.1070.0311.

Belenzon, Sharon, Aaron K. Chatterji, and Brendan Daley. 2017. "Eponymous Entrepreneurs." *American Economic Review* 107(6):1638–55. doi:10.1257/aer.20141524.

Botelho, Tristan L., Daniel Fehder, and Yael Hochberg. 2021. *Innovation-Driven Entrepreneurship*. *Working Paper*. 28990. National Bureau of Economic Research. doi:10.3386/w28990.

Botelho, Tristan L., Ranjay Gulati, and Olav Sorenson. 2024. "The Sociology of Entrepreneurship Revisited." *Annual Review of Sociology* 50(Volume 50, 2024):341–64. doi:10.1146/annurev-soc-030222-014049.

Burt, Ronald S. 2002. "Bridge Decay." *Social Networks* 24(4):333–63. doi:10.1016/S0378-8733(02)00017-5.

Burton, M. Diane, and Christine M. Beckman. 2007. “Leaving a Legacy: Position Imprints and Successor Turnover in Young Firms.” *American Sociological Review*. doi:10.1177/000312240707200206.
Campero, Santiago, and Aleksandra (Olenka) Kacperczyk. 2020. “Asymmetric Gender Homophily in the Startup Labor Market.” Pp. 329–59 in *Employee Inter- and Intra-Firm Mobility*. Vol. 41, *Advances in Strategic Management*, edited by D. Tzabbar and B. Cirillo. Emerald Publishing Limited.
Cardon, Melissa S., and Christopher E. Stevens. 2004. “Managing Human Resources in Small Organizations: What Do We Know?” *Human Resource Management Review* 14(3):295–323. doi:10.1016/j.hrmr.2004.06.001.
Carnahan, Seth. 2017. “Blocked But Not Tackled: Who Founds New Firms When Rivals Dissolve?” *Strategic Management Journal* 38(11):2189–2212. doi:10.1002/smj.2653.
Carnahan, Seth, and Brad N. Greenwood. 2018. “Managers’ Political Beliefs and Gender Inequality among Subordinates: Does His Ideology Matter More Than Hers?” *Administrative Science Quarterly* 63(2):287–322. doi:10.1177/0001839217708780.
Castilla, Emilio J. 2005. “Social Networks and Employee Performance in a Call Center.” *American Journal of Sociology* 110(5):1243–83. doi:10.1086/427319.
Chen, Jing, and Peter Thompson. 2015. “New Firm Performance and the Replacement of Founder-CEOs.” *Strategic Entrepreneurship Journal* 9(3):243–62. doi:10.1002/sej.1203.
Choi, Joonkyu, Nathan Goldschlag, John Haltiwanger, and J. Daniel Kim. 2023. “Early Joiners and Startup Performance.” *The Review of Economics and Statistics* 1–46. doi:10.1162/rest_a_01386.
Chow, Melissa C., Teresa C. Fort, Christopher Goetz, Nathan Goldschlag, James Lawrence, Elisabeth Ruth Perlman, Martha Stinson, and T. Kirk White. 2021. “Redesigning the Longitudinal Business Database.”
Chozick, Amy. 2020. “This Is the Guy Who’s Taking Away the Likes.” *The New York Times*, January 17.
Chung, Seung Hoon D., and Simon C. Parker. 2023. “Founder Affiliations: Jobseeker Reactions and Impact on Employee Recruitment by Start-up Ventures.” *Small Business Economics* 61(1):259–83. doi:10.1007/s11187-022-00694-7.
Dahl, Michael S., and Olav Sorenson. 2009. “The Embedded Entrepreneur.” *European Management Review* 6(3):172–81. doi:10.1057/emr.2009.14.
Ding, Waverly W. 2011. “The Impact of Founders’ Professional-Education Background on the Adoption of Open Science by For-Profit Biotechnology Firms.” *Management Science* 57(2):257–73. doi:10.1287/mnsc.1100.1278.
Eesley, Charles E., David H. Hsu, and Edward B. Roberts. 2014. “The Contingent Effects of Top Management Teams on Venture Performance: Aligning Founding Team Composition with Innovation Strategy and Commercialization Environment.” *Strategic Management Journal* 35(12):1798–1817. doi:10.1002/smj.2183.
Ewens, Michael, and Matt Marx. 2018. “Founder Replacement and Startup Performance.” *The Review of Financial Studies* 31(4):1532–65. doi:10.1093/rfs/hhx130.
Ewens, Michael, and Denis Sosyura. 2023. “Irreplaceable Venture Capitalists.” *Working Paper*.
Fairlie, Robert W., and Javier Miranda. 2017. “Taking the Leap: The Determinants of Entrepreneurs Hiring Their First Employee.” *Journal of Economics & Management Strategy* 26(1):3–34. doi:10.1111/jems.12176.
Freeman, John, Glenn R. Carroll, and Michael T. Hannan. 1983. “The Liability of Newness: Age Dependence in Organizational Death Rates.” *American Sociological Review* 48(5):692–710. doi:10.2307/2094928.
Ganco, Martin, Florence Honoré, and Joseph Raffiee. 2019. “Entrepreneurial Team Assembly: Knowledge Transfer, Customer Transfer, and Matching Models.” P. 0 in *The Oxford Handbook of Entrepreneurship and Collaboration*, edited by J. J. Reuer, S. F. Matusik, and J. Jones. Oxford University Press.
Goldschlag, Nathan, and Javier Miranda. 2020. “Business Dynamics Statistics of High Tech Industries.” *Journal of Economics & Management Strategy* 29(1):3–30. doi:10.1111/jems.12334.

Gompers, Paul A., Will Gornall, Steven N. Kaplan, and Ilya A. Strebulaev. 2020. “How Do Venture Capitalists Make Decisions?” *Journal of Financial Economics* 135(1):169–90. doi:10.1016/j.jfineco.2019.06.011.
Gompers, Paul, Anna Kovner, Josh Lerner, and David Scharfstein. 2010. “Performance Persistence in Entrepreneurship.” *Journal of Financial Economics* 96(1):18–32. doi:10.1016/j.jfineco.2009.11.001.
Guzman, Jorge, and Scott Stern. 2020. “The State of American Entrepreneurship: New Estimates of the Quantity and Quality of Entrepreneurship for 32 US States, 1988–2014.” *American Economic Journal: Economic Policy* 12(4):212–43. doi:10.1257/pol.20170498.
Hannan, Michael T., and John Freeman. 1984. “Structural Inertia and Organizational Change.” *American Sociological Review* 49(2):149–64. doi:10.2307/2095567.
Honoré, Florence, and Martin Ganco. 2023. “Entrepreneurial Teams’ Acquisition of Talent: Evidence from Technology Manufacturing Industries Using a Two-Sided Approach.” *Strategic Management Journal* 44(1):141–70. doi:10.1002/smj.3127.
Jaravel, Xavier, Neviana Petkova, and Alex Bell. 2018. “Team-Specific Capital and Innovation.” *American Economic Review* 108(4–5):1034–73. doi:10.1257/aer.20151184.
Jovanovic, Boyan. 1979. “Job Matching and the Theory of Turnover.” *Journal of Political Economy* 87(5, Part 1):972–90. doi:10.1086/260808.
Kacperczyk, Aleksandra J. 2013. “Social Influence and Entrepreneurship: The Effect of University Peers on Entrepreneurial Entry.” *Organization Science* 24(3):664–83. doi:10.1287/orsc.1120.0773.
Kaplan, Steven N., Berk A. Sensoy, and Per Strömberg. 2009. “Should Investors Bet on the Jockey or the Horse? Evidence from the Evolution of Firms from Early Business Plans to Public Companies.” *The Journal of Finance* 64(1):75–115. doi:10.1111/j.1540-6261.2008.01429.x.
Karaevli, Ayse, and Edward J. Zajac. 2013. “When Do Outsider CEOs Generate Strategic Change? The Enabling Role of Corporate Stability.” *Journal of Management Studies* 50(7):1267–94. doi:10.1111/joms.12046.
Keith, Kristen, and Abagail McWilliams. 1995. “The Wage Effects of Cumulative Job Mobility.” *ILR Review* 49(1):121–37. doi:10.1177/001979399504900108.
Kerr, Sari Pekkala, and William R. Kerr. 2016. “Immigrant Entrepreneurship.” *Measuring Entrepreneurial Businesses: Current Knowledge and Challenges* 187–249.
Kim, J. Daniel. 2018. “Is There a Startup Wage Premium? Evidence from MIT Graduates.” *Research Policy* 47(3):637–49. doi:10.1016/j.respol.2018.01.010.
Kim, J. Daniel. 2024. “Startup Acquisitions as a Hiring Strategy: Turnover Differences Between Acquired and Regular Hires.” *Strategy Science* 9(2):118–34. doi:10.1287/stsc.2022.0026.
Kim, Minjae, and Roberto M. Fernandez. 2023. “What Makes Weak Ties Strong?” *Annual Review of Sociology* 49(1):null. doi:10.1146/annurev-soc-030921-034152.
King, Brayden G. 2015. “Organizational Actors, Character, and Selznick’s Theory of Organizations.” Pp. 149–74 in *Institutions and Ideals: Philip Selznick’s Legacy for Organizational Studies*. Vol. 44, *Research in the Sociology of Organizations*. Emerald Group Publishing Limited.
Kleinbaum, Adam M. 2012. “Organizational Misfits and the Origins of Brokerage in Intrafirm Networks.” *Administrative Science Quarterly* 57(3):407–52. doi:10.1177/0001839212461141.
Kleinbaum, Adam M. 2018. “Reorganization and Tie Decay Choices.” *Management Science* 64(5):2219–37. doi:10.1287/mnsc.2016.2705.
Kulchina, Elena, and Pernille Gjerløv-Juel. 2019. “Entrepreneur—A Jockey or a Horse Owner?”
Lee, Joon Mahn, Dalee Yoon, and Steven Boivie. 2020. “Founder CEO Succession: The Role of CEO Organizational Identification.” *Academy of Management Journal* 63(1):224–45. doi:10.5465/amj.2017.0091.
Lyttelton, Thomas, Lasse Henriksen, and Olav Sorenson. 2024. “Social Capital and Hiring in Startups.” *Working Paper*.

Macmillan, Ian C., Robin Siegel, and P. N. Subba Narasimha. 1985. “Criteria Used by Venture Capitalists to Evaluate New Venture Proposals.” *Journal of Business Venturing* 1(1):119–28. doi:10.1016/0883-9026(85)90011-4.
Marsden, Peter V., and Karen E. Campbell. 1984. “Measuring Tie Strength*.” *Social Forces* 63(2):482–501. doi:10.1093/sf/63.2.482.
Marsden, Peter V., and Karen E. Campbell. 2012. “Reflections on Conceptualizing and Measuring Tie Strength.” *Social Forces* 91(1):17–23. doi:10.1093/sf/sos112.
Marx, Matt, and Bram Timmermans. 2017. “Hiring Molecules, Not Atoms: Comobility and Wages.” *Organization Science* 28(6):1115–33. doi:10.1287/orsc.2017.1155.
McPherson, Miller, Lynn Smith-Lovin, and James M. Cook. 2001. “Birds of a Feather: Homophily in Social Networks.” *Annual Review of Sociology* 27(1):415–44. doi:10.1146/annurev.soc.27.1.415.
O’Reilly, Charles A., Jennifer Chatman, and David F. Caldwell. 1991. “People and Organizational Culture: A Profile Comparison Approach to Assessing Person-Organization Fit.” *Academy of Management Journal* 34(3):487–516. doi:10.5465/256404.
Pillemer, Julianna, and Nancy P. Rothbard. 2018. “Friends Without Benefits: Understanding the Dark Sides of Workplace Friendship.” *Academy of Management Review* 43(4):635–60. doi:10.5465/amr.2016.0309.
Quigley, Timothy J., and Donald C. Hambrick. 2012. “When the Former Ceo Stays on as Board Chair: Effects on Successor Discretion, Strategic Change, and Performance.” *Strategic Management Journal* 33(7):834–59. doi:10.1002/smj.1945.
Roach, Michael, and Henry Sauermann. 2015. “Founder or Joiner? The Role of Preferences and Context in Shaping Different Entrepreneurial Interests.” *Management Science* 61(9):2160–84. doi:10.1287/mnsc.2014.2100.
Rocha, Vera, and Rhett Andrew Brymer. 2024. “We Go Way Back: Affiliation-Based Hiring and Young Firm Performance.” *Strategic Management Journal* n/a(n/a). doi:10.1002/smj.3673.
Ruef, Martin. 2010. *The Entrepreneurial Group: Social Identities, Relations, and Collective Action*. First Edition. Princeton: Princeton University Press.
Ruef, Martin, Howard E. Aldrich, and Nancy M. Carter. 2003. “The Structure of Founding Teams: Homophily, Strong Ties, and Isolation among U.S. Entrepreneurs.” *American Sociological Review* 68(2):195–222. doi:10.2307/1519766.
Schneider, Benjamin. 1987. “The People Make the Place.” *Personnel Psychology* 40(3):437–53. doi:10.1111/j.1744-6570.1987.tb00609.x.
Schumpeter, Joseph Alois. 1976. *Capitalism, Socialism and Democracy*. Routledge.
Scott, W. Richard. 1995. *Institutions and Organizations*. 1st edition. Thousand Oaks: SAGE Publications, Inc.
Selznick, Philip. 1957. *Leadership in Administration: A Sociological Interpretation*. First Edition. Berkeley, Cal.: University of California Press.
Selznick, Philip. 1996. “Institutionalism ‘Old’ and ‘New.’” *Administrative Science Quarterly* 41(2):270–77. doi:10.2307/2393719.
Shane, Scott, and Toby Stuart. 2002. “Organizational Endowments and the Performance of University Start-Ups.” *Management Science* 48(1):154–70. doi:10.1287/mnsc.48.1.154.14280.
Shi, Wei, Robert E. Hoskisson, and Yan Anthea Zhang. 2017. “Independent Director Death and CEO Acquisitiveness: Build an Empire or Pursue a Quiet Life?” *Strategic Management Journal* 38(3):780–92. doi:10.1002/smj.2514.
Snellman, Kaisa, and Peter Younkin. 2021. “Who’s the Boss? Evidence of Job Seeker Bias from a Field Experiment.”
Sørensen, Jesper B., and Magali A. Fassiotto. 2011. “Organizations as Fonts of Entrepreneurship.” *Organization Science* 22(5):1322–31. doi:10.1287/orsc.1100.0622.
Sorenson, Olav, Michael S. Dahl, Rodrigo Canales, and M. Diane Burton. 2021. “Do Startup Employees Earn More in the Long Run?” *Organization Science* 32(3):587–604. doi:10.1287/orsc.2020.1371.

Srivastava, Sameer B., Amir Goldberg, V. Govind Manian, and Christopher Potts. 2018. “Enculturation Trajectories: Language, Cultural Adaptation, and Individual Outcomes in Organizations.” *Management Science* 64(3):1348–64. doi:10.1287/mnsc.2016.2671.

Stinchcombe, Arthur. 1965. “Social Structure and Organizations.” in *Handbook of Organizations*. Chicago: Rand McNally.

Stinchcombe, Arthur L. 1997. “On the Virtues of the Old Institutionalism.” *Annual Review of Sociology* 23(1):1–18. doi:10.1146/annurev.soc.23.1.1.

Stuart, Toby E., and Waverly W. Ding. 2006. “When Do Scientists Become Entrepreneurs? The Social Structural Antecedents of Commercial Activity in the Academic Life Sciences.” *American Journal of Sociology* 112(1):97–144. doi:10.1086/502691.

Valentine, Jacob, and David Waguespack. 2025. “Approaches to Causal Identification in Studies of Entrepreneurship and Innovation.” Pp. 313–27 in *De Gruyter Handbook of Sociology of Innovation and Entrepreneurship*. De Gruyter.

Wasserman, Noam. 2003. “Founder-CEO Succession and the Paradox of Entrepreneurial Success.” *Organization Science* 14(2):149–72. doi:10.1287/orsc.14.2.149.14995.

Wasserman, Noam. 2012. *The Founder’s Dilemmas: Anticipating and Avoiding the Pitfalls That Can Sink a Startup*. First Edition. Princeton, New Jersey Oxford: Princeton University Press.

Weber, Max. 1947. *The Theory of Social and Economic Organization*. Mansfield Centre, CT: Martino Fine Books.

Westphal, James D., and James W. Fredrickson. 2001. “Who Directs Strategic Change? Director Experience, the Selection of New CEOs, and Change in Corporate Strategy.” *Strategic Management Journal* 22(12):1113–37. doi:10.1002/smj.205.

Whyte, William H. 1956. *The Organization Man*. Philadelphia: University of Pennsylvania Press.

Wiersema, Margarethe F. 1992. “Strategic Consequences of Executive Succession Within Diversified Firms.” *Journal of Management Studies* 29(1):73–94. doi:10.1111/j.1467-6486.1992.tb00653.x.

Yalcinkaya, Beril, Waverly W. Ding, and Anil Gupta. 2025. “Functional Scope of Capability Development in Young Firms.”

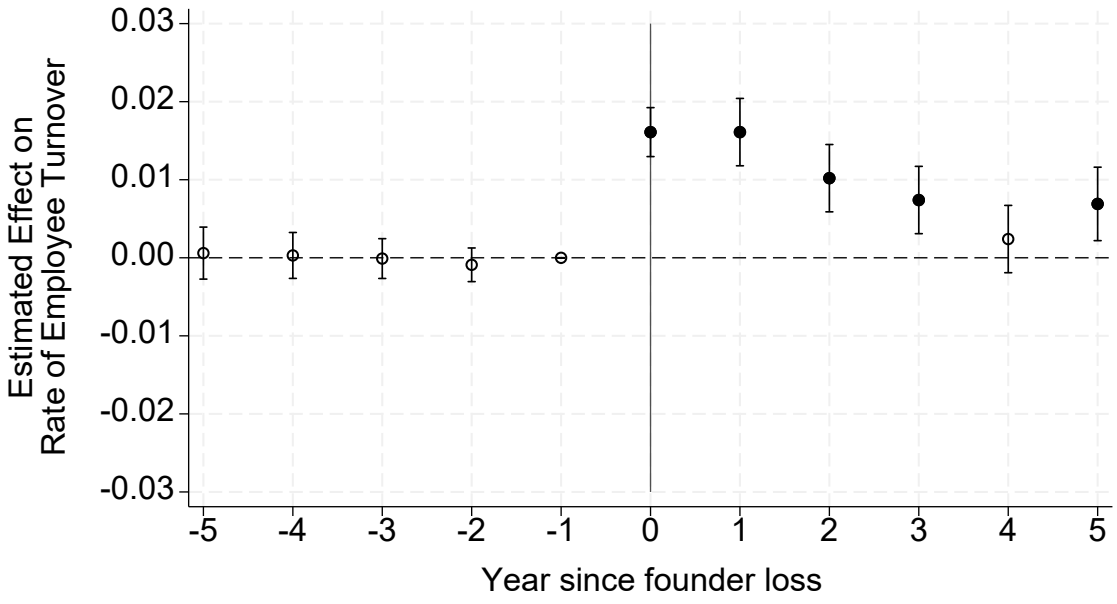
Zenger, Todd R. 1994. “Explaining Organizational Diseconomies of Scale in R&D: Agency Problems and the Allocation of Engineering Talent, Ideas, and Effort by Firm Size.” *Management Science* 40(6):708–29. doi:10.1287/mnsc.40.6.708.

Zhang, Evelyn Y., Brandy L. Aven, and Adam M. Kleinbaum. 2024. “License to Broker: How Mobility Eliminates Gender Gaps in Network Advantage.” *Administrative Science Quarterly* 00018392231221070. doi:10.1177/00018392231221070.

Zuzul, Tiona, and Mary Tripsas. 2020. “Start-up Inertia versus Flexibility: The Role of Founder Identity in a Nascent Industry.” *Administrative Science Quarterly* 65(2):395–433. doi:10.1177/0001839219843486.

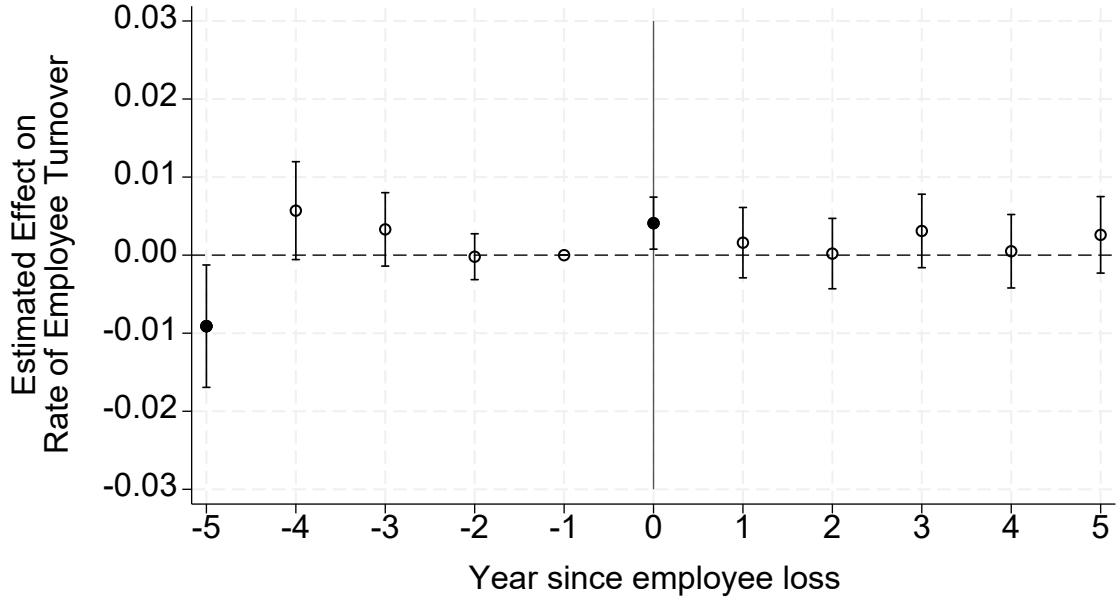
FIGURES AND TABLES

Figure 1: Event Study of Founder Loss and Employee Turnover Rates



Notes: This figure shows the event study of the estimated effect of founder loss on the startup's rate of turnover. Each of the five years before and after the founder loss is separately estimated using year *t-1* as the reference group. Sample consists of startups that lose a founder due to a premature death along with a matched control group with a placebo treatment year. For the control group, a placebo year is used. Estimated effects for the treated group are plotted along with the corresponding 95% confidence intervals. Regression specification includes year, industry (NAICS-4), state, and firm age (at time of death shock) fixed effects. Standard errors are clustered at the firm level.

Figure 2: Placebo Test: Event Study of Later Joiner Loss and Employee Turnover Rates



Notes: This figure shows the event study of the estimated effect of losing a later joiner on the startup's rate of turnover. Later joiner is defined as an employee who begins working for the startup in the second year of its operations; by definition, this individual is not a founder. Each of the five years before and after the founder loss is separately estimated using year *t-1* as the reference group. Sample consists of startups that lose a later joiner due to a premature death along with a matched control group with a placebo treatment year. For the control group, a placebo year is used. Estimated effects for the treated group are plotted along with the corresponding 95% confidence intervals. Regression specification includes year, industry (NAICS-4), state, and firm age (at time of death shock) fixed effects. Standard errors are clustered at the firm level.

Table 1: Summary Statistics of Firms

Variable Name	**Treated** (N=16,000) Mean	**Control** (N=16,000) Mean	Difference in Means	p-value
Firm Size	6.9	6.9	-0.06	0.65
Firm Age	3.2	3.2	-0.05	0.17
Payroll ($ thousands)	190	210	20	0.003
Revenues ($ thousands)	940	960	20	0.27
High-Tech	6.7%	6.7%	0.001	0.78
Rate of Employee Turnover	4.9%	4.5%	-0.004	<0.001
Rate of Employee Entrepreneurship	1.5%	1.7%	0.002	0.044

Notes: This table presents summary statistics of startups that experience the loss of a founder due to premature death, along with their matched control group. Differences in means between the two groups and the associated p-values are shown. Statistics are based on the year prior to the (placebo) death shock year. *High-Tech* is a binary variable equaling 1 if the firm's NAICS-4 industry is categorized as having a high share of STEM-oriented workers as measured by Goldschlag and Miranda (2020). Observation counts are rounded in accordance with the data provider's disclosure rules.

Table 2: Broad Sample Analysis of Founder Separations and Employee Turnover Rates

Dependent Variable	Rate of Employee Turnover		
	(1)	(2)	(3)
Post	0.018***	0.017***	0.018***
	(0.000)	(0.000)	(0.000)
Year FEs	YES	YES	YES
Industry FEs		YES	YES
State FEs		YES	YES
Firm Age FEs			YES
Observations	997,000	997,000	997,000
R-squared	0.008	0.015	0.016

Notes: This table presents regression estimates of founder separation on the startup's rate of employee turnover. Sample consists of all startups in the Longitudinal Business Database (LBD) that experience a separation of the founder before the firm age of ten. Separation is defined as an event during which an employee no longer earnings wages from his employer for whatever reason (e.g., ranging from voluntary departure or layoffs). *Post* is a binary equaling 1 for the years following founder separation. Firm Age fixed effects represent the firm's age during the year of founder separation. Industry fixed effects are based on the NAICS-4 codes. Standard errors shown in parentheses are clustered at the firm level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 3: Difference-in-Differences Analysis of Founder Loss on Employee Turnover Rates

Dependent Variable	Rate of Employee Turnover			
	(1)	(2)	(3)	(4)
Treated × Post	0.009***	0.008***	0.008***	0.015***
	(0.001)	(0.001)	(0.001)	(0.001)
Treated	0.001*	0.001*	0.001*	
	(0.001)	(0.001)	(0.001)	
Post	0.020***	0.020***	0.021***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Year FEs	YES	YES	YES	YES
Industry FEs		YES	YES	
State FEs		YES	YES	
Firm Age FEs			YES	
Firm FEs				YES
Observations	221,000	221,000	221,000	221,000
R-squared	0.015	0.032	0.032	0.253

Notes: This table presents regression estimates of founder loss on the startup's rate of employee turnover. Sample consists of startups that experience the loss of a founder due to premature death, along with their matched control group. *Treated* is a binary variable equaling 1 if the startup experiences a premature death of a founder, and 0 otherwise. *Post* is a binary equaling 1 for the years following founder loss, and 0 otherwise; for the control group, this variable is based on the placebo death shock year. Firm Age fixed effects represent the firm's age during the year of founder separation. Industry fixed effects are based on the NAICS-4 codes. Standard errors shown in parentheses are clustered at the firm level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 4: Moderating Effect of Firm Age on Founder Loss and Employee Turnover Rates

Dependent Variable	Rate of Employee Turnover			
	Split Sample		**Interaction**	
Firm Maturity Measure	Firm Age<=3	Firm Age>3	Firm Age	Firm Size
	(1)	(2)	(3)	(4)
Treated × Post	0.006***	0.012***	0.007***	0.006***
	(0.001)	(0.002)	(0.002)	(0.002)
Treated	0.001	0.002*	0.00001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Post	0.032***	0.007***	0.037***	0.026***
	(0.001)	(0.001)	(0.001)	(0.001)
Treated × Post × Firm Maturity			0.0006*	0.0003**
			(0.0003)	(0.0001)
Treated × Firm Maturity			0.0002	-0.00005
			(0.0002)	(0.0001)
Post × Firm Maturity			-0.004***	-0.001***
			(0.0002)	(0.0001)
Firm Maturity			0.002***	0.001***
			(0.0002)	(0.0001)
Year FEs	YES	YES	YES	YES
Industry FEs	YES	YES	YES	YES
State FEs	YES	YES	YES	YES
Observations	119,000	102,000	221,000	221,000
R-squared	0.039	0.032	0.034	0.036

Notes: This table presents regression estimates of founder loss on the startup's rate of employee turnover. Sample consists of startups that experience the loss of a founder due to premature death, along with their matched control group. Specifications 1 and 2 present split-sample analysis based on the firm age at which the founder loss occurs—specifically, under and above three years. Specification 3 and 4 include the full sample and instead include firm age and size as a continuous variable, respectively. All other details are identical to those in Table 3. Standard errors shown in parentheses are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1

Table 5: Summary Statistics of Individuals

Variable Name	**Treated,** N=129,000 Mean	**Control,** N=128,000 Mean	Diff-in-Means	p-value
Annual Earnings ($2010)	31,600	33,000	1400	<.001
Age	38.3	38.1	-0.2	0.01
Male	58%	56%	0.023	<.001
Foreign born	15%	19%	-0.035	<.001
Tenure (in years)	2.4	2.3	0.1	<.001

Notes: This table presents summary statistics of individuals whose employers experience founder loss due to premature death, along with their counterparts at matched firms. Differences in means and their associated p-value are presented. Sample conditions on employees who are present in the year of the founder loss; in other words, employees, who leave the startup prior to the year of the founder loss are excluded. *Tenure* is the years of employment shared between the individual and the founder at the startup until the loss of the founder; for the control group, this measure is based on the placebo death shock year. Annual earnings ($2010) and age are measured based on the year of the founder loss.

Table 6: Cox Hazard Regressions of Employee Turnover

Dependent Variable	Employee Turnover				
	(1)	(2)	(3)	(4)	(5)
Treated	0.131***	0.045***	0.042***	0.088***	0.087***
	(0.003)	(0.005)	(0.005)	(0.008)	(0.008)
Tenure		-0.095***	-0.091***	-0.087***	-0.082***
		(0.001)	(0.001)	(0.001)	(0.001)
Treated × Tenure		0.033***	0.033***	0.039***	0.040***
		(0.001)	(0.001)	(0.002)	(0.002)
Age			-0.003***	-0.004***	-0.003***
			(0.000)	(0.000)	(0.000)
Male			0.014***	-0.046***	-0.031***
			(0.004)	(0.005)	(0.006)
Foreign born			-0.075***	-0.044***	-0.076***
			(0.005)	(0.007)	(0.007)
Year FEs				YES	YES
Industry FEs					YES
State FEs					YES
Observations	258,000	258,000	258,000	258,000	258,000
pseudo R-squared	0.000182	0.00183	0.00197	0.0197	0.0209

Notes: This table shows Cox hazard regressions estimating individuals' likelihood of turnover. *Treated* is a binary variable equaling 1 if the individual's startup experiences a founder loss due to premature death, and 0 otherwise. *Tenure* is the years of employment shared between the individual and the founder at the startup until the loss of the founder; for the control group, this measure is based on the placebo death shock year. Coefficients, rather than hazard ratios, are shown. Robust standard errors are shown in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 7: Cox Hazard Regressions of Employee Turnover by Shared Sex

Dependent Variable	Employee Turnover			
Cox Hazard Model	(1)	(2)	(3)	(4)
Treated	0.089***	0.083***	0.117***	0.109***
	(0.006)	(0.006)	(0.009)	(0.009)
Same Sex	-0.020***	-0.009*	-0.053***	-0.040***
	(0.005)	(0.005)	(0.008)	(0.008)
Treated × Same Sex		0.056***	0.095***	0.109***
		(0.008)	(0.011)	(0.011)
Tenure		-0.074***	-0.067***	-0.062***
		(0.001)	(0.001)	(0.001)
Age		-0.003***	-0.004***	-0.003***
		(0.000)	(0.000)	(0.000)
Foreign born		-0.074***	-0.044***	-0.077***
		(0.005)	(0.007)	(0.007)
Year FEs			YES	YES
Industry FEs				YES
State FEs				YES
Observations	258,000	258,000	258,000	258,000
pseudo R-squared	0.00193	0.00192	0.0196	0.0209

Notes: This table shows Cox hazard regressions estimating individuals' likelihood of turnover. Only signs, rather than the magnitudes, of the estimates are shown due to Census disclosure rules. *Treated* is a binary variable equaling 1 if the individual's startup experiences a founder loss due to premature death, and 0 otherwise. *Same Sex* is a binary variable equaling 1 if the individual and the underlying founder are both female or both male, and 0 otherwise. *** p<0.01, ** p<0.05, * p<0.1

Table 8: Placebo Test of Later Joiner Loss and Employee Turnover Rates

Dependent Variable	Rate of Employee Turnover			
	(1)	(2)	(3)	(4)
Treated × Post	0.001	0.001	0.001	0.002
	(0.001)	(0.001)	(0.001)	(0.002)
Treated	0.001	0.001	0.001	
	(0.001)	(0.001)	(0.001)	
Post	0.044***	0.045***	0.045***	0.026***
	(0.001)	(0.001)	(0.001)	(0.001)
Year FEs	YES	YES	YES	YES
Industry FEs		YES	YES	
State FEs		YES	YES	
Firm Age FEs			YES	
Firm FEs				YES
Observations	113,000	113,000	113,000	113,000
R-squared	0.046	0.091	0.092	0.317

Notes: This table presents regression estimates of losing a later joiner—defined as an individual whose employment begins in the second year of the startup's operations—on the startup's rate of employee turnover. Sample consists of startups that experience the loss of a later joiner due to premature death, along with their matched control group. *Treated* is a binary variable equaling 1 if the startup experiences a premature death of a later joiner, and 0 otherwise. *Post* is a binary equaling 1 for the years following later joiner loss, and 0 otherwise; for the control group, this variable is based on the placebo death shock year. Firm Age fixed effects represent the firm's age during the year of founder separation. Industry fixed effects are based on the NAICS-4 codes. Standard errors shown in parentheses are clustered at the firm level. *** p<0.01, ** p<0.05, * p<0.1